

Aayush Gadamshetty

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science, Minor in Cybersecurity (GPA: 3.5)

College Station, TX

Aug. 2025 – May 2027

University of Texas at Dallas

Bachelor of Science in Computer Science (GPA: 4.0)

Richardson, TX

Aug. 2024 – May 2025

TECHNICAL SKILLS

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda), Azure DevOps, Infrastructure as Code (IaC), CI/CD, Scrum/Agile.

Languages: Python (Scripting Proficiency), Java, C++, R, SQL, HTML/CSS/JS, Linux/Bash, Assembly.

Libraries/Tools: Pandas, NumPy, Scrapy, Matplotlib, Git/GitHub, VS Code, AutoCAD.

EXPERIENCE

Undergraduate Research Assistant (Computational Biology)

Aug. 2025 – Present

Texas A&M University

College Station, TX

- Architected automated **Python** pipelines to process high-dimensional scRNA-seq data, managing datasets of **5000+** colorectal cells for cancer marker identification.
- Engineered distributed computing workflows to quantify gene expression heterogeneity, utilizing **UMAP** and **t-SNE** to visualize complex cell populations.
- Optimized data preprocessing scripts in **Python/R**, reducing computational overhead by **25%** for large-scale genetic analysis.
- Maintained rigorous technical documentation and version control to ensure reproducibility across research teams.

Backend Development Specialist (Cloud Services)

Sep. 2024 – Jun. 2025

Quila

Dallas, TX

- Developed backend microservices for **AI Assistant** and **Analytics** modules using **Python** and **AWS** infrastructure.
- Maintained and optimized **APIs**, implementing performance monitoring that identified bottlenecks and improved response times.
- Collaborated within a **Scrum** framework, participating in daily stand-ups and sprints to ensure seamless data integration across cloud applications.
- Managed security protocols by maintaining **IAM** roles and permissions for development and testing environments.

Systems Support & IT Operations Intern

Jun. 2023 – Oct. 2023

Quick MD Care

Frisco, TX

- Administered clinic **IT systems**, handling user access requests and ensuring 99.9% uptime for critical healthcare software.
- Conducted routine system audits and troubleshooting to resolve hardware and network connectivity issues for medical staff.
- Drafted technical documentation for software workflows, enhancing administrative efficiency and data privacy compliance.

PROJECTS

Algorithm Trading Indicator | Python, AWS, Statistical Modeling, Pine Script

Sept. 2024 – Jan. 2025

- Developed a mid-frequency trading engine in **Python** that yielded a **32.59%** return on XAU/USD through automated back-testing.
- Achieved a **1.824** profit factor and optimized risk management scripts to maintain a low maximum drawdown of **5.84%**.
- Utilized cloud-based statistical modeling to outperform buy-and-hold benchmarks by **4.52%** with a **0.952** Sharpe ratio.

COURSEWORK

Machine Learning (CS 421), Artificial Intelligence (CS 420), Analysis of Algorithms (CS 411), Software Engineering (CS 331), Computer Systems (CS 313), Data Structures (CS 221), Cybersecurity Minor.